

THE INNOVATOR'S DILEMMA

Revolutionary Thinking that will Change
the Way you do Business

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THE BIG "SO WHAT"

Why do good, successful companies that do all the "right" things still fail or lose their market leadership in face of unexpected competition? The book answers this question, and shows how **the same (good) practices that lead to a business's success can eventually lead to its demise – this is the innovator's dilemma.** The book proposes a set of rules that CEOs, entrepreneurs and managers can apply to solve this dilemma.



HOW CAN GREAT FIRMS FAIL?

Insights from the Hard Drive Industry

Christensen started his research using the hard disk drive industry, which is **characterized by rampant and rapid changes, with a high rate of failure:**

- Between the 1970s to 1990s, the price per megabyte (MB) decreased at about 5% per quarter, while the amount of information in one square inch of disk surface increased by 35% per year on average.
- Between 1976 and 1995, all of the 17 (relatively large) firms in the industry - except IBM - failed or were acquired. Another 109 out of the 129 new entrants also failed.

Did these firms fail because they could not cope with the relentless onslaught of technology change? No, **they failed because of the introduction of "disruptive technologies" in their industry.**



In particular, the 2 management principles below are important to leading firms' success. However, they are also the reason for their demise in the face of unexpected technology change:

KEY QUOTES

"This book is about the failure of...well-managed companies that have their competitive antennae up, listen astutely to their customers, invest aggressively in new technologies, and yet still lose market dominance."

An Innovator's Dilemma
"The logical, competent decisions of management that are critical to the success of their companies are also the reasons why they lose their positions of leadership."

"Many of what are now widely accepted principles of good management are, in fact, only situationally appropriate."

"Blindly following the maxim that good managers should keep close to their customers can sometimes be a fatal mistake."

"Neither the pace nor the difficulty of technological change lay at the root of the leading firms' failures."

- Always listen to and respond to the needs of your best customers.
- Focus your investments on innovations that promise the highest returns

The book explains this dilemma in detail.



SUSTAINING TECHNOLOGIES VS DISRUPTIVE TECHNOLOGIES

Christensen differentiates between 2 strategically different technologies. Established firms generally beat newcomers at sustaining innovations, but lose their leadership positions due to disruptive innovations.

[Note: Technology in the book refers to “the process by which an organization transforms labor, capital, materials and information into products and services of greater value”. Innovation refers to a change in one of these technologies].

 Sustaining Technology	 Disruptive Technology
 Improve performance of established products	 Underperform compared to established products
 Meet demands of mainstream customers in major markets	 Can't serve mainstream markets but have new features that are valued by new/ fringe customers
 Led by established firms, who have the resources and setup to constantly invest in innovations.	 Cheaper, simpler, smaller, more convenient to use
	 Led by entrant firms

Disruptive innovations are often technologically straightforward, e.g. PCs vs mainframes. They are usually a novel combination of off-the shelf components that are cheaper, simpler, smaller, and more convenient to use. They cannot serve the needs of mainstream markets, but offer value to fringe consumers and hence create new markets (that are initially unattractive to incumbents). Over time, they can be improved to take over older markets.



BARRIERS TO INNOVATION

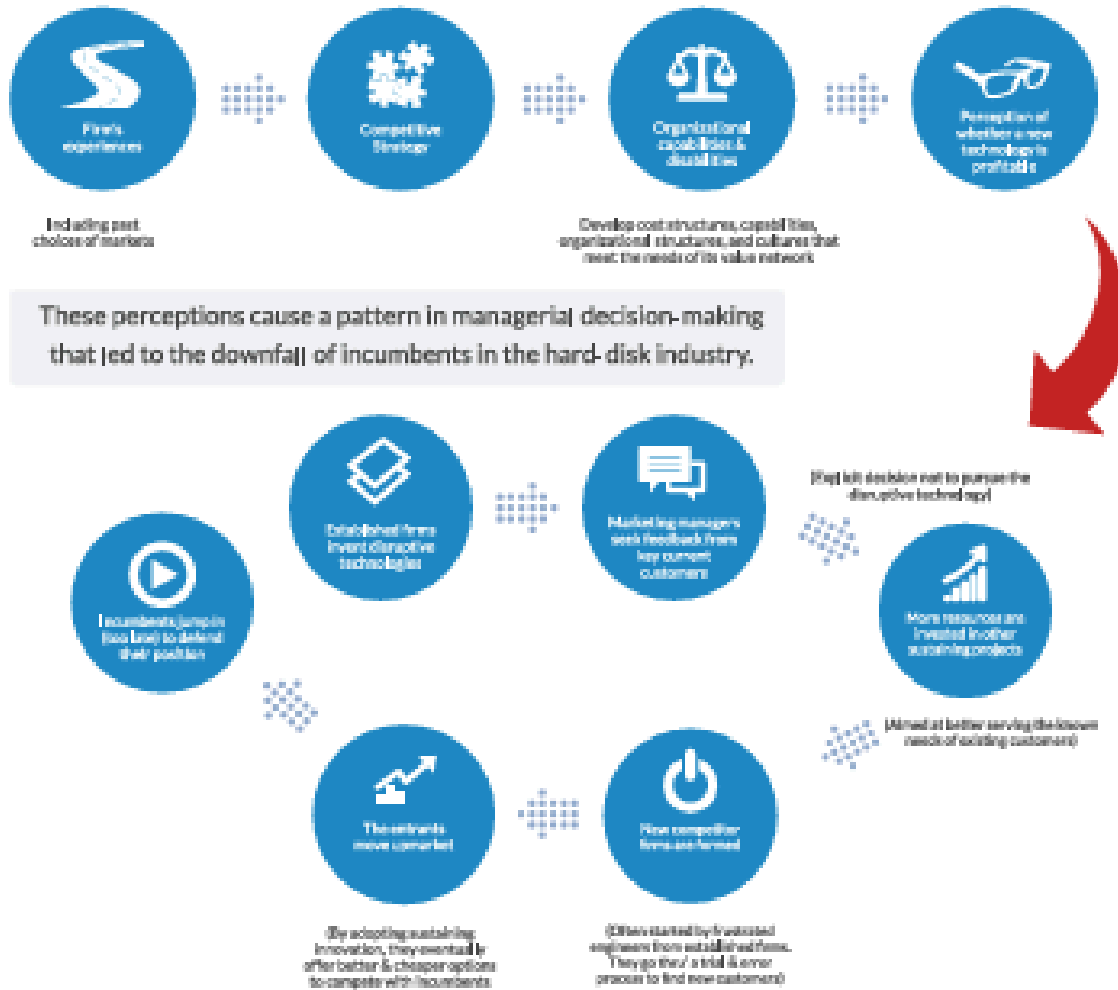
Through his study, Christensen found that leading firms do not commit to disruptive technologies until it is too late. There are 2 factors at play:

KEY QUOTES

“Their failures resulted from delay in making the strategic commitment to enter the emerging market.”



Value Networks: These are the contexts within which firms identify and respond to customers' needs, solve problems, secure resources, react to competitors and seek profits. A company's past experiences results in its perceptions and eventually decisions/patterns of resource allocation within the value network.



The **technology S-curve** is commonly used by established firms for decision-making on how to invest in or improve their solutions for existing customers, based on the technology lifecycle. Disruptive technologies pose a challenge because they cannot be plotted and analyzed on the same chart as the typical S-curves (as they offer different types of value).



Barriers to downward mobility: Established firms also face 3 barriers to downward mobility:



(i) Characteristic cost structures

Most leading companies are successful because they have developed their image, cost and profit structures that allow them to be competitive in their value networks. It is hard to cut costs to move down-market profitably.

KEY QUOTES



(ii) Resource allocation

Decision-making models (whether top-down or bottom-up) tend to favor upward mobility in resource allocation. Management tends to select projects that are most consistent with the firm's strategy, offer the highest return on investment, and has the most assured market demand.



(iii) Upmarket movement of customers

The firm's customers may themselves be migrating upmarket, prompting the firm to invest in offerings with higher/more premium value.

As leading firms continue to move up-market, it creates a vacuum in the lower-end markets, attracting firms with lower cost structures and simpler technologies to enter.



THE IMPACT OF DISRUPTIVE INNOVATION IN OTHER INDUSTRIES

Christensen showed how the Innovator's Dilemma also affected firms in the Mechanical Excavator industry and the Steel industry:



The Mechanical Excavator Industry had leading firms with a long history of successful innovations, moving from steam-powered to gasoline-powered excavators. However, the incumbents were wiped out because they did not embrace hydraulic excavators until it was too late. Younger entrants that focused on this disruptive innovation eventually dominated the market (though it took 20 years for the hydraulic technology to become viable for mainstream customers).



Steel Industry. "Minimills" started out operating at a fraction of the cost and scale of large "integrated mills", serving different market niches. Because of the small scale and different operating models, they could embrace new technologies and opportunities in a way that the larger integrated mills could not, and gradually moved upmarket to replace the integrated mills.

KEY QUOTES



THE 5 PRINCIPLES OF DISRUPTIVE INNOVATION

To address the innovator's dilemma, Christensen proposes 5 laws/principles of disruptive technology, derived from studying a mix of companies that succeeded and failed when faced with disruptive technologies. He stresses the importance for managers to understand these principles rather than to use them as model answers.



Principle 1:

Companies depend on customers & investors for resources

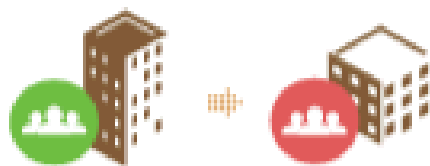
The theory of resource dependence says that customers and investors (not managers) are the ones who really control how resources are allocated in organizations.

Good resource allocation processes are hence designed to sieve out proposals that customers don't want, by default weeding out disruptive innovations. Managers also typically focus on:

- (a) Projects that are aligned with the company's most profitable customers and products; and
- (b) Highly profitable innovation programs that can propel their personal career forward.

Disruptive innovations do not fulfil these considerations.

Solution: Create an independent organization that is responsible to serve the new customer needs using disruptive technologies.



The same organization cannot concurrently support mainstream customers and pursue disruptive

innovations, due to the different cost structures and models required. From the case studies, it is seen that mainstream companies that successfully established a strong position in disruptive technologies did so by **setting up new and independent businesses around the technologies.**

This places the technology in a different value network, allows managers to use different success metrics, think like new entrants to the markets, and be free from current constraints.

The book provides detailed case studies on the success/failure of:

KEY QUOTES

The 5 Laws

"I'm particularly anxious that managers read these chapters for understanding, rather than for simple answers."

"It is the customers, rather than the managers, who really determine what a firm will do. It is forces outside the organization, rather than the managers within it, that dictate the company's course."

"Projects targeted at the explicit needs of current customers or at the needs of existing users...will always win over proposals to develop products for markets that do not exist. The best resource allocation systems are designed precisely to weed out ideas that are unlikely to find large, profitable, receptive markets."

"Creating an independent organization, with a cost structure honed to achieve profitability at the low margins characteristic of the most disruptive technologies, is the only viable way for established firms to harness this principle."

- Digital Equipment Corporation (DEC), IBM and the personal computer
- Kresge, Woolworth and discount retailing
- Hewlett-Packard's Laser-jet vs Ink-jet printers



Principle 2:

Small markets don't solve the growth needs of large companies

Large and successful companies typically don't invest in disruptive technologies because:

- Although disruptive technologies usually create new markets, these new markets are initially not attractive enough to large companies. For example, a 20% growth for a \$40 million company is just \$8m, while a 20% growth for a \$4 billion company is \$800 million – no new market is singularly large enough for large companies.
- Disruptive technologies are typically simpler and cheaper, with lower profit margins. They also tend to be most suitable for launch with new and less profitable customers in the market.

Although it is viable to adopt a follower strategy in either sustaining or disruptive technologies, a leadership position in disruptive technologies can bring rapid growth and enormous value in the long run.

Solution: Match the size of the organization to the size of the market.



When setting up an independent organization, match its size to that of the market. This ensures that:

- The smaller organization can respond most nimbly to the new market(s), to develop first-mover advantages without distracting the leaders from the main business of the company; and
- The market is large enough to excite the management of the new organization(s).

Johnson & Johnson, for example, has 160 autonomous companies, each of which can introduce small disruptive products (e.g. disposable contact lenses).

KEY QUOTES

"Small organizations can most easily respond to the opportunities for growth in a small market."



Principle 3:

Markets that don't exist can't be analyzed

Sustaining technologies work well with the process of sound market research, good planning, followed by execution. However, for disruptive technologies, the markets are new and there's no existing market data. None of the past market data, revenues or costs projections are relevant.

Solution: *Match the size of the organization to the size of the market.*



When doing "discovery-based planning", managers should work on the assumption that their forecasts and strategies are wrong.

Investments and management focus should be on learning what needs to be known.

The initial learning is largely through trial-and-error, and failures are inevitable. Take a flexible and exploratory approach in your strategy, and set aside adequate resources to revise your approach based on real-time learning.



Marketing strategy:

- Openly acknowledge that the disruptive technology is not currently applicable to mainstream markets. Instead, focus on creating a market that could appreciate its attributes, such that its weaknesses become its strengths.
- Avoid using existing research data for decision-making. Instead, use "agnostic marketing", i.e. watch how customers use the products, rather than listen to what they say. Assume that no one – even the customers – know if, how, or in what quantities they will use the product until they try it.
- Set up a business plan for learning (not executing a fixed strategy). Identify the most critical information needed, and in what sequence the information is required.



Product Development Strategy:

Develop a product that:

- Is simple, reliable, and convenient;

KEY QUOTES

"Not only are the market applications for disruptive technologies unknown at the time of their development, they are unknowable."

"Because markets for disruptive technologies are unpredictable, companies' initial strategies for entering these markets will generally be wrong."

"Guessing the right strategy at the outset isn't nearly as important to success as conserving enough resources...so that new business initiatives get a second or third stab at getting it right."

"(Successful new entrants) sold their products without a clear marketing strategy—essentially selling to whoever would buy. Out of what was largely a trial-and-error approach to the market, the ultimately dominant applications for their products emerged."

- Has features, functions and styles that can be made quickly and at low cost; and
- Must hit a low price point

Technology Strategy:

Disruptive technologies usually offer great value by putting together proven technologies in a novel way. Technological breakthroughs are hence not required.

Distribution Strategy:

Acknowledge the need to find new distribution channels aligned with the new market requirements. The book provides detailed case studies on:

- Hewlett-Packard and the failed Kittyhawk project; and
- Honda's successful invasion of the North American motorcycle industry

Principle 4: *An organization's capabilities define its disabilities.*

Good managers tend to assign capable people to the job, based on their skills, experiences and values. **Just because an individual has the requisite skills for the job doesn't mean the organization has the requisite capabilities.**

Solution: Appraise your organization's capabilities and disabilities.

To identify where the core competencies of a firm reside, Christensen recommends the Resources-Processes-Values (RPV) framework.

Resources:

These are assets or things "that can be hired and fired, bought and sold, depreciated or enhanced", e.g. people, equipment, technology, product designs, brands, information, energy, cash, relationships. They usually can be transferred.

Processes:

These are the methods by which the organization transforms inputs of resources into outputs of higher value. They are visible, and can be formal, informal or cultural in nature. These processes help employees to perform recurrent tasks consistently but can

KEY QUOTES

"Organizations have capabilities that exist independently of people who work within them."

"To succeed consistently, good managers need to be skilled not just in choosing, training and motivating the right people for the right job, but in choosing, building and preparing the right organization for the job as well."

backfire when applied to disruptive technologies.

 Values:

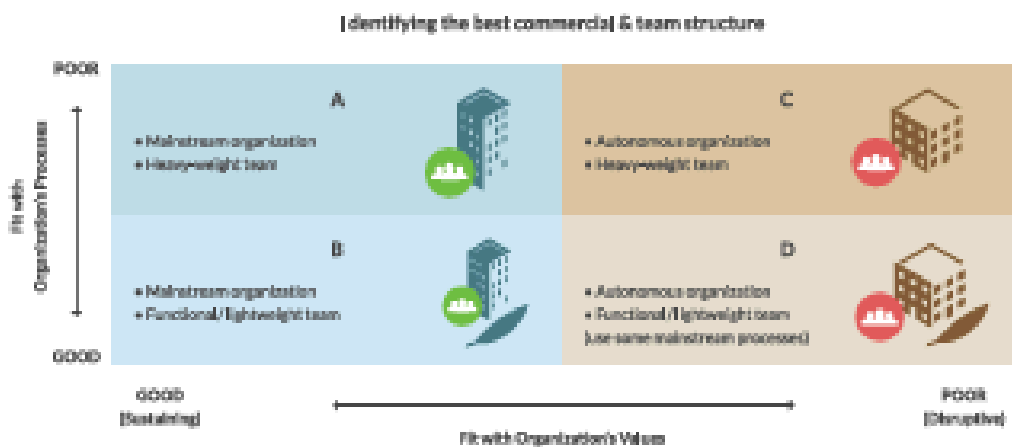
These are the criteria that determine how prioritization decisions are made. Successful organizations train employees at every level to make independent decisions that are aligned with the company's strategic direction and business model. These same values, however, also define what an organization can't do, especially with regard to the acceptable levels of profits and size of businesses.

Leading firms succeed in adopting sustaining technology because they have the resources, rigorous processes and investments that fit the companies' values. However, **disruptive innovations happen so rarely that no firm has a routine process to manage them, and the margins typically do not fit with the company values.**

To prepare the organization to succeed in the disruptive technology, **managers must go beyond resource allocation, to also scrutinize the organization's processes and values.** There are **3 ways to create new capabilities:**

-  Acquisition of a different organization with processes and values suited for the new task
-  Internal creation by changing current processes and values
-  Creation of independent organizations to develop new processes and values

Using the framework below, managers can select the best commercial structure and development team for the challenge. Generally, functional/lightweight teams are good for exploiting established capabilities, and heavyweight teams are good for creating new ones.



KEY QUOTES

"Clear, consistent, and broadly understood values, however, also define what an organization cannot do."

"When the organization's capabilities reside primarily in its people, changing to address new problems is relatively simple. But when the capabilities have come to reside in processes and values and especially when they have become embedded in culture, change can become extraordinarily difficult."



Principle 5:

Technologies can progress faster than market demand

Because technology investment and development tends to outstrip the growth in market demand, large companies tend to move up-market too quickly, over-satisfying the needs of their original customers ("performance oversupply"). In the process, they secure higher-margin/higher-performing markets, but create a vacuum at the lower price points that allow competitors with competitive technologies to come in. These technologies become disruptive when they (eventually) become fully performance-competitive against established products.

Solution: Manage the evolution of product competition, and learn to spot disruptive threats & opportunities

According to Windermere Associates, customers make their choice between competing products in the market using this "buying hierarchy": **Functionality => Reliability => Convenience => Price**. Companies can choose between 3 strategic options when managing the evolution of product competition:



To identify if a new technology is potentially disruptive, define current mainstream market needs by watching what they do, not simply by listening to what they say. **Compare the current market needs with the new technology capabilities, to check if:**



The new technology works on fundamentally different value assumptions (hence they can't be used in the mainstream markets today and are likely to be doubted by existing market leaders); but



Could one day improve to be competitive in parts of the mainstream market.

KEY QUOTES

"Differentiation loses its meaning when the features and functionality have exceeded what the market demands."